Lock-Based Protocols & Multiple Granularity DBMS

> **Definition: Lock-Based Protocols** control concurrent access to data items by requiring transactions to acquire **Locks** (Shared or Exclusive) before reading or writing data.

---

1. Detailed Technical Explanation

1. Lock Modes:

1. **Shared Lock (S-Lock):** Granted for `READ` operations. Multiple transactions can hold S-locks simultaneously on the same item.

2. **Ex
read o

Lock Compatibility Matrix:

| Lock

---

2. Two-Phase Locking Protocol (2PL)

2PL guarantees conflict serializability by dividing a transaction's lock lifecycle into two distinct phases:

```

Numbe

### 2PL Variants:

1. **Basic 2PL:** Transaction acquires locks as needed (Growing), releases locks (Shrinking). Can suffer from **Cascading Aborts**.

2. **St

---

## 3. Multiple Granularity Locking (MGL)

MGL allows data items of various sizes (Database -> File -> Page -> Record) to be locked in a tree hierarchy.

```

[DATA

Intent Locks:

Before locking a fine-grained node (e.g., Record), a transaction must acquire an **Intent Lock** on its ancestor nodes:

- **IS

---

4. Core Concepts & Memory Keywords

- **Lock Point:** Point in time where a transaction acquires its final lock in 2PL.

- **C

---

5. Must-Write Points for Exams

- Basic 2PL guarantees **conflict serializability**, but does NOT prevent **deadlocks**.

- Stri

---

6. Quick Recall Flow

```

Growi
```